Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



07/07/2026

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2025-26 (ODD)	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	22	Name of Student	Sanskruti . P . Kharapkar

Practical Number	8(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a Typed To-Do List application using TypeScript and React, demonstrating interfaces, typed functions, components, props, and the `useState` hook.
Problem Definition	To create an interactive To-Do List application that allows users to add tasks, mark them as completed, and delete them. The application should use TypeScript interfaces to define task data, typed functions to manage tasks, and React components to display and update the user interface dynamically.
Theory (100 words)	TypeScript is a programming language that adds static typing to JavaScript and helps identify type-related errors during development. An interface defines the structure of an object, while typed functions specify parameter and return types. React is a JavaScript library used to create user interfaces through reusable components. Props allow data and functions to be passed from parent components to child components.

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Procedure and Execution

(100 Words)

Step for Implementation:

1. **Create a React + TypeScript project using Vite.**
2. **Install the required dependencies using `npm install`.**
3. **Create the `types.ts` file and define the Task interface.**
4. **Implement typed task functions in `todoLogic.ts`.**
5. **Create `TaskInput.tsx` and `TaskItem.tsx` components.**
6. **Update `App.tsx` to manage tasks using `useState`.**
7. **Apply styling through `App.css` and `index.css`.**
8. **Run the application and test all task operations.**

Code:

- **types.ts:** Defines the Task interface with `id`, `title`, and `completed` properties.

```
export interface Task {
    id: number;
    title: string;
    completed: boolean;
}
```

- **todoLogic.ts:** Contains typed functions to add, complete, delete, and retrieve tasks.

```
import type { Task } from "./types";

let tasks: Task[] = [];

export function addTask(title: string): void {
    const newTask: Task = {
```

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```
        id: Date.now(),
        title: title,
        completed: false
    };

    tasks.push(newTask);
}

export function toggleTask(id: number): void {
    tasks = tasks.map((task: Task) => {
        if (task.id === id) {
            return {
                ...task,
                completed: !task.completed
            };
        }

        return task;
    });
}
```

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```
export function deleteTask(id: number): void {

    tasks = tasks.filter(

        (task: Task) => task.id !== id

    );

}



export function getTasks(): Task[] {

    return [...tasks];

}
```

- **App.tsx:** Manages task state and renders the task list using React components.

```
import { useState } from "react";

import TaskInput from "./components/TaskInput";

import TaskItem from "./components/TaskItem";

import type { Task } from "./types";

import "./App.css";



function App() {

    const [tasks, setTasks] = useState<Task[]>([]);
```

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```
    function addTask(title: string): void {

        const newTask: Task = {

            id: Date.now(),

            title: title,

            completed: false

        };



        setTasks((previousTasks) => [...previousTasks,
newTask]);

    }



    function toggleTask(id: number): void {

        setTasks((previousTasks) =>

            previousTasks.map((task) =>

                task.id === id

                    ? { ...task, completed:
!task.completed }

                    : task

            )

        );

    }
```

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```
    function deleteTask(id: number): void {
        setTasks((previousTasks) =>
            previousTasks.filter((task) => task.id !==
id)
        );
    }

    return (
        <div className="todo-app">
            <h1>Typed To-Do List</h1>

            <TaskInput onAdd={addTask} />

            {tasks.length === 0 ? (
                <p>No tasks yet. Add your first task!</p>
            ) : (
                <ul>
                    {tasks.map((task) => (
                        <TaskItem
                            key={task.id}
                            task={task}
                            onToggle={toggleTask}
```

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```
                                onDelete={deleteTask}

                            />

                        ))}

                    </ul>

                )}



                <p>Total tasks: {tasks.length}</p>

            </div>

        );

}



export default App;
```

- **main.tsx:** Renders the main React application.

```
import { StrictMode } from "react";

import { createRoot } from "react-dom/client";

import "./index.css";

import App from "./App";



createRoot(document.getElementById("root")!).render(
```

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```
    <StrictMode>
        <App />
    </StrictMode>
);
```

- **App.css and index.css:** Provide styling for the user interface.

```
import { useState } from "react";
import TaskInput from "./components/TaskInput";
import TaskItem from "./components/TaskItem";
import type { Task } from "./types";
import "./App.css";

function App() {
    const [tasks, setTasks] = useState<Task[]>([]);

    function addTask(title: string): void {
        const newTask: Task = {
            id: Date.now(),
            title: title,
            completed: false
```

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```
        };



        setTasks((previousTasks) => [...previousTasks,
newTask]);

    }



    function toggleTask(id: number): void {

        setTasks((previousTasks) =>

            previousTasks.map((task) =>

                task.id === id

                    ? { ...task, completed:
!task.completed }

                    : task

            )

        );

    }



    function deleteTask(id: number): void {

        setTasks((previousTasks) =>

            previousTasks.filter((task) => task.id !==
id)

        );
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```
    }


    return (
        <div className="todo-app">
            <h1>Typed To-Do List</h1>

            <TaskInput onAdd={addTask} />

            {tasks.length === 0 ? (
                <p>No tasks yet. Add your first task!</p>
            ) : (
                <ul>
                    {tasks.map((task) => (
                        <TaskItem
                            key={task.id}
                            task={task}
                            onToggle={toggleTask}
                            onDelete={deleteTask}
                        />
                    ))}
                </ul>
```

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```
            )}


            <p>Total tasks: {tasks.length}</p>

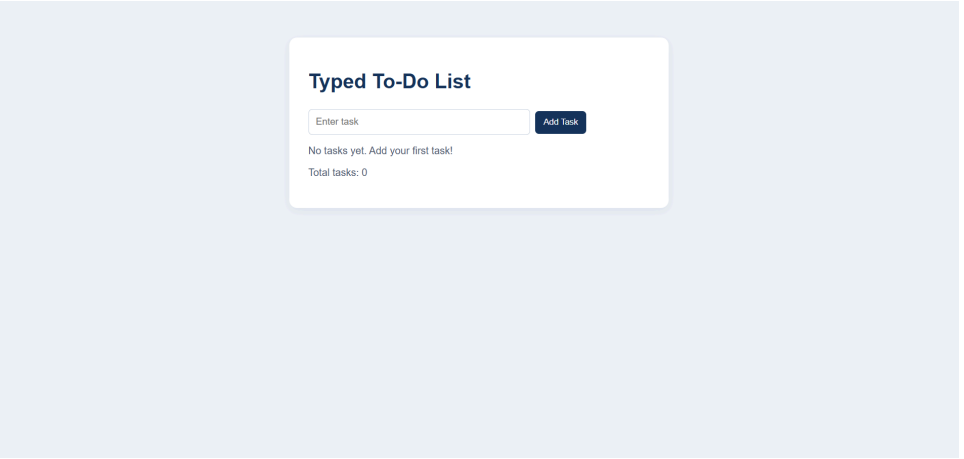
        </div>

    );

}


export default App;
```

Output:

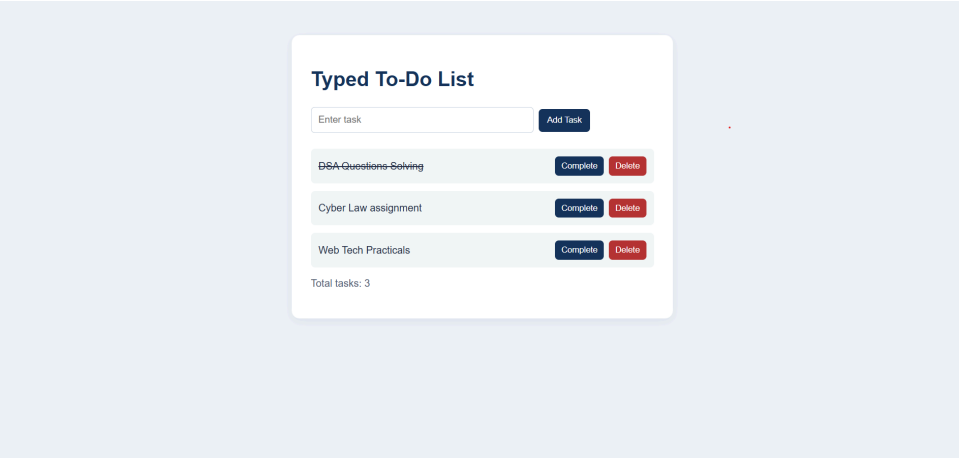


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Output Analysis	**The application demonstrates TypeScript interfaces, typed functions, React components, props, and state management. The Task interface ensures a consistent structure for task objects. The `useState` hook updates the interface when the task list changes. The `map()` method renders task components, while `filter()` removes selected tasks. Event handlers connect user actions to task operations. The application successfully demonstrates the integration of TypeScript and React for building an interactive user interface.**
Link of student Github profile where lab assignment has been uploaded	**https://github.com/Sanskruti2007-code/web-Technology-Practicals**
Conclusion	**The Typed To-Do List application demonstrates the use of TypeScript and React to develop an interactive task management system. It provides functionality for adding, completing, and deleting tasks. Through this practical, the concepts of interfaces, typed functions, components, props, `useState`, and array methods are applied. The project improves understanding of type safety, reusable components, and dynamic user interface development.**
Plag Report (Similarity index < 12%)	**Similarity index 10%**

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Date	**06/10/26**